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MILITARY SPECIFICATION

FURNACES, HEAT TREATING, ELECTRIC, BATH TYPE

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bath type, electrically heated, heat treating furnaces of-the round and rectangular-pot design with equipment and accessories specified in section 3. The requirements set forth in this specification and the completed ordering data (paragraph 6.2) establish the Government's minimum requirements for each procurement.

1.2 Classification. The furnaces shall be of the following types, classes and of the sizes listed in Tables I, II, III, IV and V. The types, classes and sizes to be furnished shall be as specified (see 3.5 and 6.2).

Type I- Round pgt, External Resistance Heated
  Class A - 600 F to 1650 F operating temperature

Type 11 - Rectangular Pot, electrode Nested
  Class A - 600 F to 1200 F operating temperature
  Class B - 1200° F to 1700° F operating temperature
  class c - 1700°F to 2400°F operating temperature

Type 111 - Rectangular Pot, Immersion Resistance Heated
  Class A - 350 F to 750 F operating temperature

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3424

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings. and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with a specific acquisition should be obtained from the contracting activity.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the isssues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI MC96.1 - Temperature Measurement Thermocouples.

(Applications for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NENA)

NENA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA ICS 3 - Industrial Systems. (DOD adopted)

NEMA ICS 6 - Enclosures for Industrial Controls and Systems. (DOD adopted)

NEMA MG1 -Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1680.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 86 - Ovens & Furnaces. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4), in accordance with 4.4.

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3.2 Design, The furnace shall be new (not a prototype), and of the manufacturer's current production models capable of operations in accordance with the requirements herein. The furnace shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement or repair. The design and construction of the machine shall be in accordance with either the U.S. Customary System of Units (US) or the International System of Units (SI).

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the furnace shall be graduated in the same system.

3.2.2 Type I. The Type I furnace shall be of the round pot design, externally resistance heated. The heating elements shall surround the pot and shall be located on the insulation wall in the space between the pot wall and Insulation.

3.2.3 Type II. The Type II furnace shall be of the rectangular pot design, internally heated by either suspended immersion-type electrodes, or internally heated by built-in submerged electrodes.

3.2.4 Type III. The Type III furnaces shall be of the rectangular pot design internally heated-by immersion resistance elements. When required, the Type III furnace shall be provided with quench pumps, heat exchangers, and agitation devices for quenching processes. Quench equipment and performance requirements shall be as specified (see 6.2).

3.2.5 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.6 Energy efficiency. The furnace and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.7 Controls. All mechanical and electrical operating controls shall be located convenient to the operator's work station(s).

3.2.8 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the furnace, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial furnace supplied to the commercial market shall be provided for the point of operation and

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all other parts of the furnace that present safety hazards. If point-of-operation guarding is influenced by workplace size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). Al 1 special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not Interfere with the operation of the furnace. All guards shall allow access to the guarded part(s).

3.2.9 Environmental protection. The furnace shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. During the operation of the furnace at its final location. it shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract regulating the emission of materials hazardous to the environment. Specific environmental protection requirements exceeding the established EPA and OSHA regulations as required herein shall be as specified by the procuring activity (see 6.2).

3.2.10 Lubrication Where necessary to reduce wear or maintain efficiency, lubrication shall be provided in a manner so as to function without replacement for not less than the manufacture's published service intervals. Areas requiring manual lubrication shall be readily accessible or shall be provided with a device that will allow for the replacement or addition of required lubricants. All other areas shall be prodded with readily accessible grease fittings of filler caps as required by application. The unit shall be initially charged with the manufacturer’s recommended lubrication as specified on the lubrication chart required in paragraph 3.9.1. Lubrication applications and systems shall be designed to prevent any contamination of furnace atmosphere, components, material or equipment.

3.2.10.1 Lubricants. Any greases, oils, or fluids used to initially lubricate the furnace shall be free of contaminants known to be hazardous to human health.

3.2.11 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the Interchangeability of respective replacement parts between end items of the same model without modification of the part or the furnace. All replacement parts shall be available domestically.

3.3 Construction. The furnace shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding temperature variations encountered during operation of the furnace to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging. or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

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3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Melding, brazing, or soldering shall be employed only where specified in the original design. Any excess material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the furnace and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard' section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The furnace shall consist of not less than a shell, drain, cover, pot , transformer(s), all accessory electrical motors, contractors, controls, relays, signal lights and switches prewired where possible for operation upon connection to source of power.

3.4.1 Shell. The outer shell of each Type I and Type III furnace shall be fabricated of sheet steel, reinforced and welded on all sides and the bottom. Each Type II furnace may have a ceramic outer shell. The shell thickness shall be not less than 10 gage, U.S. Standard Sheet Metal Gage, for the Type I furnace and not less than 0.188 inch for Type 11 and Type III furnace. The submerged electrode design furnace may have a ceramic outer shell. The top of the bath (excluding cover), shall be between 32 and 38 inches above floor level, except for furnaces requiring greater height for reasons of capacity. Unless otherwise specified (see 6.2), the Type II and Type 111 furnaces shall be mounted on steel I-beam skids or similar material to elevate the bottom of the furnace not less than 3 inches above the floor. The top of the cabinet shall be constructed of material to prevent distortion. Cabinet construction shall prevent contact between insulation and bath material.

3.4.2 Drain. The Type I furnace shall be provided with drainage facilities for the bath material in case of pot wall failure. The Insulation surrounding the drain area shall be sloped to the drain.

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3.4.3 Type I furnace cover. When specified (see 6.2), the Type I furnace shall be equipped with an insulated swivel-mounted, split-type cover.

3.4.4 Type II and Type III furnace cover. When specified (see 6.2), the Type II and Type 111 furnace shall be provided with a hand operated one or two piece rolling cover. The cover shall be insulated or made of castable refractory. When specified (see 6.2), the cover shall be electric motor powered.

3.4.5 Insulation. Each furnace shall be insulated with insulating fire brick, block, ceramic fiber, or pre-formed insulation selected to insure long life and minimum heat loss. Powdered, shredded or granular insulation is not acceptable. Quality and thickness of the insulation shall be such that the outside surface temperature of the furnace will not exceed the maximum surface temperature specified in 3.4.5.1, 3.4.5.2, and 3.4.5.3 for the respective type and class furnace specified.

3.4.5.1 Type I. furnace insulation. The insulation adjacent to the heating elements shall have a temperature rating of not less than 2000 degrees F and the temperature rating of the back-up insulation shall be not less than 1600 degrees F. The maximum outside surface temperature of the furnace shall not exceed 80 degrees F above ambient air temperature with the bath temperature stabilized at 1600 degrees F.

3.4.5.2 Type II, Class A and Type III, furnace insulaltion. The insulation adjacent to the steel pot shall have a temperature rating of not less than 400 degrees F above the maximum temperature of the furnace. The maximum outside surface temperature of each furnace shall not exceed 60 degrees F above ambient air temperature with the bath of the Type II furnace stabilized at 1100 degrees F and the bath of the Type III furnace stabilized at 750 degrees F.

3.4.5.3 Type II. Class B and C furnace insulation. The Type II, Class B and C furnaces, when furnished with ceramic pots, shall have back-up insulation consisting of refractory lumnite cement block or pre-formed insulation of sufficient quality and thickness to insure not more than 8 percent by volume bath material (salt) absorption during the life of the furnace, or a secondary leak proof steel pot separating the ceramic pot from the insulation. The secondary steel pot, if used, shall be not less than 3/16 of an inch thick and welded inside and out. Maximum outside surface temperature for the Type II, Class B furnace shall not exceed 80 degrees F above ambient air temperature with the bath temperature stabilized at 1600 degrees F, and 140 degrees F above ambient air temperature for the Type II, Class C furnace, with the bath temperature stabilized at 2200 degrees F.

3.4.6 Pot. Each furnace pot shall be capable of continuous operation at the maximum operating temperature specified in 1.2 for each type and class furnace without sagging or warping. Metal pots shall be not less than 3/8 inch thick and shall be free from porosity. Additional characteristics shall be in accordance with 3.4.6.1, 3.4.6.2, and 3.4.6.3 for the type and class of furnace specified.

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3.4.6.1 Type I furnace pot. The Type I pot shall be cylindrical with a rounded bottom and shall be cast from an alloy containing not less than 35 percent nickel and 18 percent chromium. Means shall be provided for properly sealing the space between the pot flange and top plate.

3.4.6.2 Type II and Type III. Class A furnace wt. Unless otherwise specified (see 6.2), the pot for the Type II and Type III, Class A furnace shall be fabricated from an aluminized low-carbon steel plate. All seams shall be welded inside and out.

3.4.6.3 Type II. Class B and C furnace pot. Unless otherwise specified (see 6.2), the pot for the Type II and Type 111, Class B and C furnace shall be fabricated from pre-formed and pre-fired salt-resistant ceramic material.

3.4.7 The furnace heating elements. Heating elements shall be of the rod or ribbon type, fabricated from an alloy of 80 percent nickel and 20 percent chromium. The elements shall be mounted on the heating chamber wall in the space between the pot wall and insulation, properly insulated from each other, and distributed in such a way to produce uniform temperature specified in 3.4.5. Heating elements shall not be located in the bottom of the heating chamber. Power density on the heating elements shall not exceed 15 watts per square inch of radiation surface area.

3.4.7.1 Type II furnace heating elements. The Type II furnace shall be heated either by suspended immersion type electrodes or shall be heated by built-in submerged electrodes. The electrodes shall be spaced and located in the bath in such a way that full transformer power may be obtained through the transformer taps. The portion of the electrode which is subject to salt corrosion shall be stainless steel.

3.4.7.2 Type III furnace heating elements. The Type III furnace shall be heated by sheath immersion resistance type elements. The elements shall enter the bath from the top and be provided with necessary protection to prevent being damaged by the work load.

3.4.8 Transformer. Type II furnaces designed to operate on single phase power shall be provided with a single phase, continuous duty, 6 tap (7 or 8 tap optional) transformer. Furnaces designed to operate on three phase power shall be provided with three phase Scott connected, continuous duty, 6 tap (7 or 8 tap optional) transformers (2 units). One designated as a "main" and the other a ‘teaser”. Each transformer shall be rated 1/2 the total KVA required, and in the event of a line fault causing a failure of one unit, either the main or the teaser can be used single phase. Unless otherwise specified (see 6.2), all transformers at any KVA rating shall be water cooled. If air cooled transformers are specified, the design shall provide for convection cooling. The design for either water or air cooling shall be adequate to prevent a temperature rise in excess of safe operating temperature for the insulation system used. Each transformer shall have a thermostat system for overload protection.

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3.4.9 Starting method. When specified (see 6.2), an electric starting unit or starting torch kit shall be provided for starting a frozen or new bath in the Type II furnace.

3.4.9.1 Electric starting unit. The electric starting unit shall consist of an electric resistance type heating unit with transformer leads.

3.4.9.2 Starting torch kit The starting torch kit shall consist of a burner, handle, extension tube, pressure gage, pressure regulator, and not less than ten feet of hose with suitable connections for connecting to commercial gas bottles.

3.4.10 Thermocuples, sensor, and target tubes Thermocouples shall conform to ANSI MC96.1 and each thermocouple and sensor shall be provided with not less than 50 feet of suitable lead wire.

3.4.10.1 Type I. and Type II. Class A and B. Each furnace shall be provided with two chromel-alumel type thermocouple assemblies. The Type I furnace shall provide for locating one thermocouple in-the heating chamber between the insulation and pot wall for the temperature controller and one thermocouple located in the bath for the excess temperature controller. The Type II, Class A and B furnace shall provide for locating two thermocouples in the bath, one for the excess temperature control and one for the temperature controller.

3.4.10.2 Type II. Class C. The Type II, Class C furnace shall be provided with two radiation-type sensors and target tubes for each furnace, one for the temperature controller, and one for the excess temperature control or high-low controller.

3.4.10.3 Type III. The Type III furnace shall be provided with one iron-constant type thermocouple assembly for locating in the bath and connecting to the temperature controller.

3.4.11 Type I furnace instrument panel. Unless otherwise specified (see 6.2), the Type I furnace shall be provided with one factory-wired, front opening floor-mounted instrument panel conforming to NEMA ICS 1 for mounting all instruments and accessory controls. All controls shall be suitably mounted within the panel. A separate enclosure shall be provided for mounting the power contractor individual and separate from the instrument panel. The panel shall provide on-off signal lights and flush mounting of all instruments.

3.4.11.1 Type 11 furnace instrument panel. Unless otherwise specified (see 6.2), the Type 11 furnace shall be provided with an instrument panel as specified for the Type I furnace. In addition to the panel, the Type II furnace shall be provided with a floor-mounted switch gear control cubicle totally enclosed and factory-wired with the switch gear components suitably mounted within the cubicle. The control cubicle shall conform to NENA ICS 6.

3.4.12 Instruments. The instruments shall be of solid state design. They shall be marked to indicate the particular type thermocouple for which it is calibrated and have automatic reference junction compensation in all instruments using

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thermocouples as the temperature sensing element. The controllers shall provide thermocouple upscale fail-safe protection that will shut off the heat source at approximately **100 F** above maximum temperature.

3.4.12.1 Type I and Type 11. The Type I and Type 11 furnaces shall be provided with a single indication-recording, controlling type instrument. Unless otherwise specified (see 6.2), the instrument shall be of the time proportioning type control and shall operate a contractor through an intermediate relay system. When specified (see 6.2), the Type II, Class C furnace shall be provided with saturable core reactor or a continuous proportional SCR control system. Either system furnished shall have an accuracy of 0.25 percent or better, of full scale reading and shall be capable of maintaining the set control temperature within plus or minus **2 F** within operating range of the furnace.

3.4.12.1.1 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type. Unless otherwise specified (see 6.2), the calibrated chart width shall be not less than six inches, and the chart length shall be not less than 80 feet to the roll. The minimum chart speed shall be one inch per hour. The chart mechanism shall be furnished in a dust-proof housing with a glass door that pemits full view of chart width, and not less than seven inches of chart 1ength.

3.4.12.1.2 Control indicator scale. The temperature control indicator scale shall be approximately the same length as the temperature recorder chart width. The calibrated scale range shall be not less than 100° F above the maximum temperature of the furnace. The increments between successive temperature marks shall be not greater than 20° F, with the spacing of numbered marks not greater than 400° F. The control indicator shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature, and one pointer shall indicate the actual furnace temperature.

3.4.12.1.3 Excess temperature controller. When specified (see 6.2), each furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the control circuitry. The controller shall provide for selecting the over-temperature control point and shall function to shut-off the heat source to the furnace when the over-temperature control point is reached. The excess temperature controller shall provide for furnace shut-down in the event of thermocouple failure, or failure of other components in the circuit. In case the furnace is shut-down by the excess temperature control system, it shall sound an audible alarm and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

3.4.12.1.4 High-Low temperature controller. When specified (see 6.2), in lieu of the excess temperature controller specified in 3.4.12.1.3, the furnace shall be provided with a High-Low temperature controller. The High-Low control circuit shall be independent of the control circuitry and the controller shall provide for selecting the high temperature control point and the low temperature control point. The High-Low controller shall interrupt the heat source to the furnace when the high temperature control point is reached and sounds an alarm. The temperature

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shall coast down to the low control point and the temperature shall be maintained at the low point and the alarm shall continue to sound.

3.4.12.2 Type III. Unless otherwise specified (see 6.2), the Type 111 furnace shall be provided with an indicating, non-recording, automatic temperature co trolling system capable of maintaining the bath temperature within plus or minus 10 F of control temperature. The control indicator shall have a calibrated scale length of not less than six inches, shall be calibrated in degrees Fahrenheit and in increments of 5° F with the spacing of numbered marks not greater than 100° F. The control indicator shall have a maximum range of not less than 100° F above the maximum range of the furnace. When specified (see 6.2), the Type III, furnace shall be provided with the temperature control recorder specified for the Type I and 11 furnace and the instrument panel specified for the Type I furnace.

3.4.13. Electrical requirements. The furnace shall be wired to conform to NFPA 86. Each furnace shall be wired to draw all of Its electrical power from a single power supply. Unless otherwise specified (see 6.2), the furnace shall be designed to operate on a single 230-volt, 60 Hertz (Hz) input power supply. Furnaces up to and including 5 KU input rating shall operate on single phase. Furnaces above 5 KW inputs, up to 40 KM input rating shall be either single phase or 3 phase. Furnaces above 40 KU input rating shall operate on 3 phase. All power actuated, manually adjusted and automatic controls shall operate on and shall be wired for 115-volt, 60-Hertz (Hz) current.

3.4.13.1 Electrical connections. When necessary to maintain effucuebct and to protect the equipment from excess heat, the terminals connecting the transformer to the electrodes shall be water cooled. Terminals of the electrical heating elements and electrodes shall have at least twice the current carrying capacity of the heating elements and electrodes, and shall be of the solderless type. Other electrical connections and terminals shall be of the pressure or solder type. When soldered connections are used, the conductors and terminal shall be mechanically secured before soldering. Soldering flux, when used, shall be of a resin base.

3.4.13.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. All motors shall conform to the requirements of NEMA MG-1. Unless otherwise specified (see 6.2), motors shall be alternating-current (AC), 60-Hertz (Hz), or direct-current (DC) industrial-type motors with dripproof enclosures. Each motor control shall conform to the requirements of NEMA ICS-1 and NEMA ICS-3.

3.5 Size. Unless otherwise specified (see 6.2), the furnace specified shall conform to one of the sizes tabulated in Tables I through V. The clear inside dimensions of the furnace shall not be less than the dimensions tabulated or otherwise specified.

3.5.1 Physical size limitations. Physical size of the furnace shall be limited when specified (see 6.2). The physical size of the furnace shall not exceed the restriction(s) specified by the procuring activity to ensure the furnace will fit in its future operating location (see 6.10).

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3.6 Performance requirements. Each furnace shall be capable of continuous operation at its maximum operating temperature. Temperature variation within the working area of the bath shall be not more than plus or minus 5 F from any preset temperature within the operating range of the furnace after the temperature has stabllized at set point. The furnace shall be of such design and construction that it will provide a stable support for the bath material and a load in the bath of not less than 125 pounds per cubic foot of bath capacity. Unless otherwise specified (see 6.2), minimum power input requirements shall be as specified in the tables herein for the type, class, and size furnace required.

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial furnace shall be furnished.

3.8 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified and fully described (see 6.2). The equipment provided shal be functional without requiring modification of the equipment or furnace.

3.9 Marking on instruments. control panels. charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.9.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each furnace. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below:

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.9.2 Nameplate. A nameplate shall be securely attached to each furnace. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Workmanship. Workmanship of the furnace and its accessories shall meet all requirements specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibllity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), al 1 Inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The furnace shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each furnace prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8.1 thru 4.8.4 and the inspection in 4.9. The furnace shall pass the examination, all tests, and the inspection to be accepted.

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4.6 Acceptance test. An acceptance test shall be performed on each furnace to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the furnace is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8.1 thru 4.8.4.2. The furnace shall pass the examination and all tests to be accepted.

4.7 Examination. Each furnace shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of **41° to 104** Fahrenheit and 10 to 95 percent relative humidity.

4.8.2 Instrument test. Indicating and recording, instruments shall be tested by impressing various predetermined voltages and simulating thermocouple outputs for different furnace temperatures to terminals where normally the thermocouples would be connected. These tests shall be made in accordance with and results checked against standards established in ANSI MC96.1. The instruments shall be tested through their operating range.

4.8.3. Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance, and faulty grounds in accordance with the requirements of NEMA ICS 6.

4.8.4 Performance test. The heat treating furnace selected for test shall be filled to 2/3 pot capacity with appropriate bath material. An indicating or recording wattmeter shall be connected in the heater power circuit to indicate the power input. With all circuits energized, the power input shall be measured at the point of maximum power consumption between start and operating temperature and shall be not less than the minimum KW input specified for the type, class, and size furnace under test. The furnace shall then be operated at the temperature specified in 3.4.5 until equilibrium has been established. The outer shell temperature shall not exceed the maximum surface temperature specified for the type and class furnace under test.

4.8.4.1 Temperature variation. Thermocouples with suitable indicating instruments shall be so located in the bath as to determine that the temperature variation specified in 3.6 has not been exceeded. This test shall be conducted at both minimum and maximum furnace operating temperature.

4.8.4.2 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.8. The

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Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.8.6 Electromagnetic Interference control test. When specified (see 6.2), equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.7 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I, class A furnaces are intended primarily to be used in lead bath operations and operations requiring salts which are not compatible with ceramic materials. The Type II, class A and Type 111, class A furnaces may be used for most salt bath operations in the lower temperature range. The Type II, class B and C furnaces are intended to be used in operations requiring neutral salts in the middle and upper temperature ranges.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class and size furnace required (see 1.2).

co Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article inspection, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound or metric) (see 3.2.1).

f. Quench equipment and performance characteristics, when required (Type III only) (see 3.2.4).

g" Additional safety and health requirements, if required (see 3.2.8).

h. Point of operation guarding, if required (3.2.8).

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j. Specify additional environmental protection requirements, if required (see 3.2.9)

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

l. Mounting skids (Type 11 and III), if not required (see 3.4.1).

m. Cover (Type I), if required (see 3.4.3).

n. Cover (Type 11 and III), if required, (specify manual or power operated), (see 3.4.4).

o. Pot material (Type II and III class A), if different, (see 3.4.6.2).

P" Pot material (Type II, class B and C), if different, (see 3.4.6.3).

q. Water cooled transformer, if required (see 3.4.8).

r. Starting device (Type II), if required (see 3.4.9).

s. Instrument panel, if different (Type I) (see 3.4.11).

t. Instrument panel and switch gear cubicle, if different (Type II) (see 3.4.11.1).

u. Time proportioning type control, if different (see 3.4.12.1).

v. Saturable core reactor or SCR controller (Type 11 class C), if required (see 3.4.12.1).

w. Chart width and length, if different (see 3.4.12.1.1).

x. Excess temperature controller, if required (3.4.12.1.3).

y. High-Low controller in lieu of excess temperature control, if required (see 3.4.12.1.4).

z. Automatic temperature control system, if different (Type 111) (see 3.4.12.2), (specify if the same instruments and panel are required as specified for Type I and 11 furnaces).

aa. Electrical requirements, if different (see 3.4.13).

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bb. Motors, if different (see 3.4.13.2).

cc. Specific inside dimensions, if non-standard size is specified (see 3.5).

dd. Physical size limitations (see 3.5.1)

ee. Power input requirements (KW), if different (see 3.6).

ff. Specify and fully describe optional and accessory equipment, if required (see 3.8).

gg. Lubrication plate, if required (see 3.9.1).

hh. Lubrication plate information, if different (see 3.9.1).

ii. Nameplate, if different (see 3.9.2).

jj. Inspection conditions, Indifferent (see 4.3).

kk. First article inspection, Indifferent (4.4).

ll. Quality conformance inspection, if different (see 4.5).

mm. Acceptance test, if different (see 4.6).

nn. Additional performance tests, if required (see 4.8.4.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-THSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (ANSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current Inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

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6.9 Hazardous materials minimization. The manufacturer should minimize the use, generation, or emission of materials known hazardous to the environment during the manufacturing processes employed at his facilities. During the production of an end item, the manufacturer should comply with all applicable Federal, state, and local statutes in effect at the point of production which regulate the use of hazardous materials. Where practical, alternative materials or processes should be employed in lieu of known hazardous materials or processes to minimize the threat to the environment.

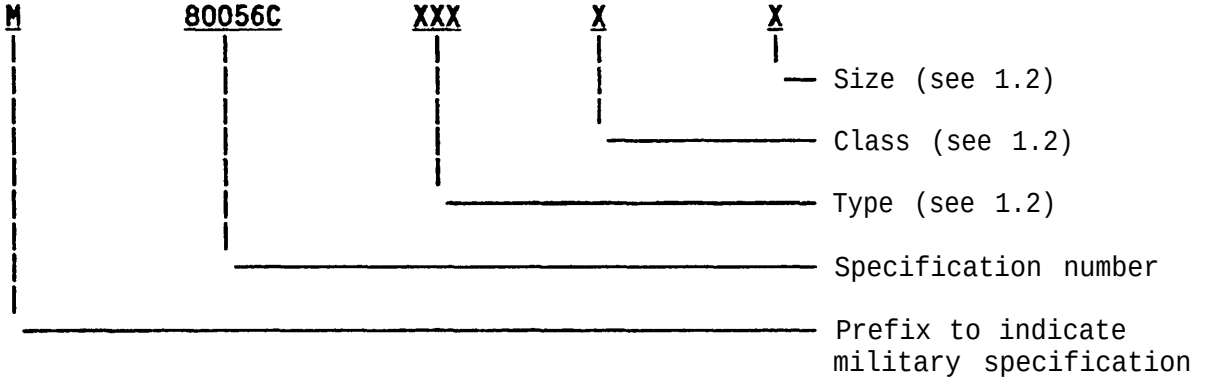
6.10 Physical size limitations. The procuring activity should be careful when restricting the physical size of the furnace so as not to create conflict between the specified interior dimensions, heating equipment requirements, insulation requirements, and exterior restriction(s).

6.11 Training. Training required by the procuring activity should be provided as specified in the contract.

6.12 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.13 Inspection 1ocation. The contractor should identify in his response to the solicitation the location where inspection and tests are to-be performed

6.14 Part or identifying Number (PIN) The PIN to be used for machines acquired to this specification are created as follows:



NOTE: Use Roman Numerals when designating furnace Type.
Use of Capital letters when designating furnace Class.

6.15 Subject term (keyword) listing.
Heat process equipment
Metal: hardening, normalizing, tempering
Atmospheres
Metallurgy

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TABLE I. Round Pot, external resistance heated
Type I - Class A

Size Furnace bath Working dimensions in inches			Input KW Not less than
Size	Diameter	Depth	
A	8	10	8
B	10	14	17
C	12	18	20
D	16	18	27
E	16	30	34
F	24	24	40

TABLE II. Rectangular pot, electrode heated
Type II - Class A

Size Furnace bath Working dimensions in inches				Input KW Not less than
Size	Length	Width	Depth	
A	12	7	22	15
B	15	9	22	18
C	18	15	22	25
D	24	15	22	32
E	36	20	24	50

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TABLE III. Rectangular pot, electrode heated
Type II - Class B

Size	Furnace bath Working dimensions in inches			Input KW Not less than
	Length	Width	Depth	
A	9	5	22	22
B	9	7	30	28
C	12	11	22	40
D	15	9	30	40
E	24	15	22	70
F	24	15	30	80

TABLE IV. Rectangular pot, electrode heated
Type 11 - Class C

Size	Furnace bath Working dimensions in inches			Input KW Not less than
	Length	Width	Depth	
A	9	7	22	35
B	12	9	22	50
C	15	9	22	60
D	18	12	22	90
E	21	15	30	90
F	24	15	30	100

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TABLE V. Rectangular Pot. immersion electrode heated
Type III - Class A

Size Furnace bath Working dimensions in inches				Input KW
Size	Length	Width	Depth	Not less than
A	12	12	24	10
B	18	12	24	10
C	24	17	24	20
D	24	17	30	20
E	36	17	24	20
F	36	17	30	20
G	48	17	24	30
H	48	17	30	30

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - SN, CR
Air Force - 84
DLA - GS

User activities:
Army - AR, SM
Navy - AS, MC, OS

Preparing activity:
DLA - IP

Project number:
(3424 - 0117)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-80056C	2. DOCUMENT DATE (YYMMDD) 91-10-10

3. DOCUMENT TITLE

FURNACES, HEAT TREATING, ELECTRIC, BATH TYPE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME NSE INDUSTRIAL PLANT EQUIPMENT CENTER ATT: DIPEC-SSM	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (901) 775-6257 DSN 683-6257
c. ADDRESS (Include Zip Code) 2163 AIRWAYS BLVD MEMPHIS, TN 38114-5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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